Variability, Standard Deviation

Variability

There is more variability if your data is distributed far from the mean, le

You can indeed have the same mean but high, or low, variability! As you
and y are not the same, yet the red line representing their respective m

```
# two different arrays
x = np.array([2,3,5,6])
y = np.array([-8,-7,15,16])

# needed for the graph below
my_range = np.arange(4)

# their respective means
mean_x = np.mean(x)
mean_y = np.mean(y)

fig, axs = plt.subplots(1,2, sharey=True)
axs[0].stem(my_range, x, label='x values', bottom
axs[1].stem(my_range, y, label='y values', bottom
legend_x = axs[0].legend(loc='upper right')
legend_y = axs[1].legend(loc='lower right')
plt.show()
```

Remember this important idea: "the average of deviations is always zer
about why that is. If you look at the above two charts, that may help you
variability of your data around the mean, then by adding all these numb
mean is, you could say, the middle of all these deviations, where they ca

We then use the square of the values to obtain more information: when
cancellation of all deviations has been avoided. Now our deviations can
calculating the variance:

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}.$$

When we are done with our calculation, and have computed the standa
are now taking the square root in our equation, after having squared ev

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}.$$